< JavaScript />

Activity06 – Variables_Operators_Arrays

You will work on three JavaScript files :

netId_Activity06_variables.js

netId_Activity06_operators.js

netId_Activity06_arrays.js

Sarfaraz Ahmed Mohammed, Ph.D.
Ali Jannesari, Ph.D.

References to online tutorials and courses

https://www.javascript.com/learn

https://www.w3schools.com/js/

https://developer.mozilla.org/en-US/

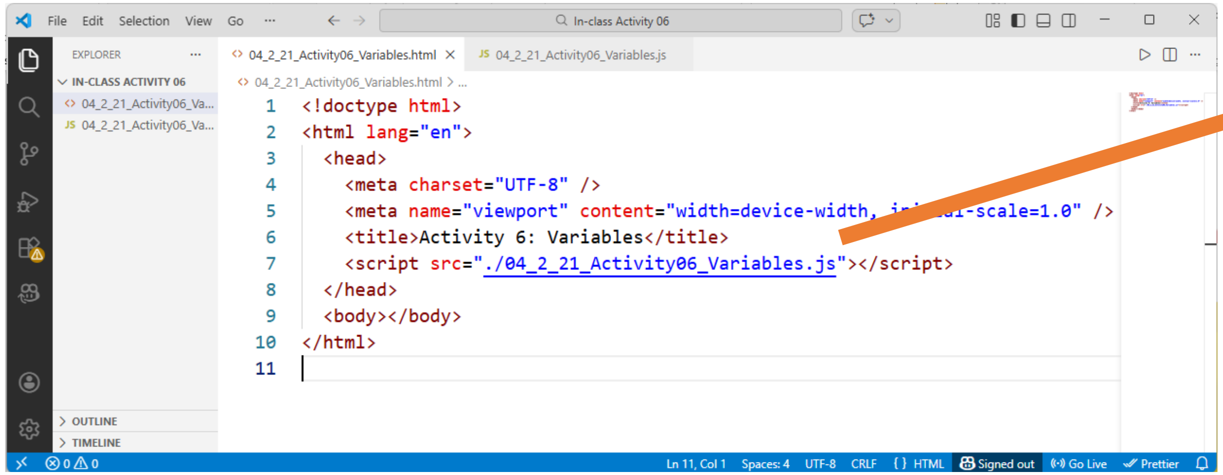
https://www.udemy.com/

Write a JavaScript:

`netId_Activity06_variables.js`

Write a HTML:

`netId_Activity06_index.html`



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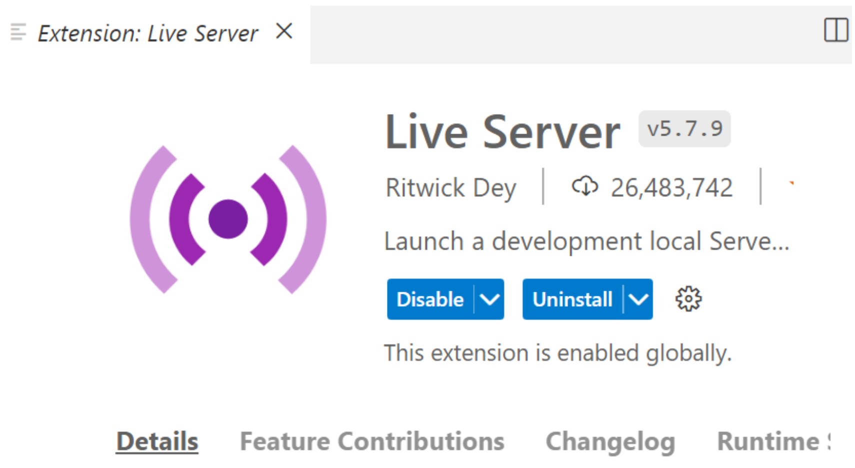
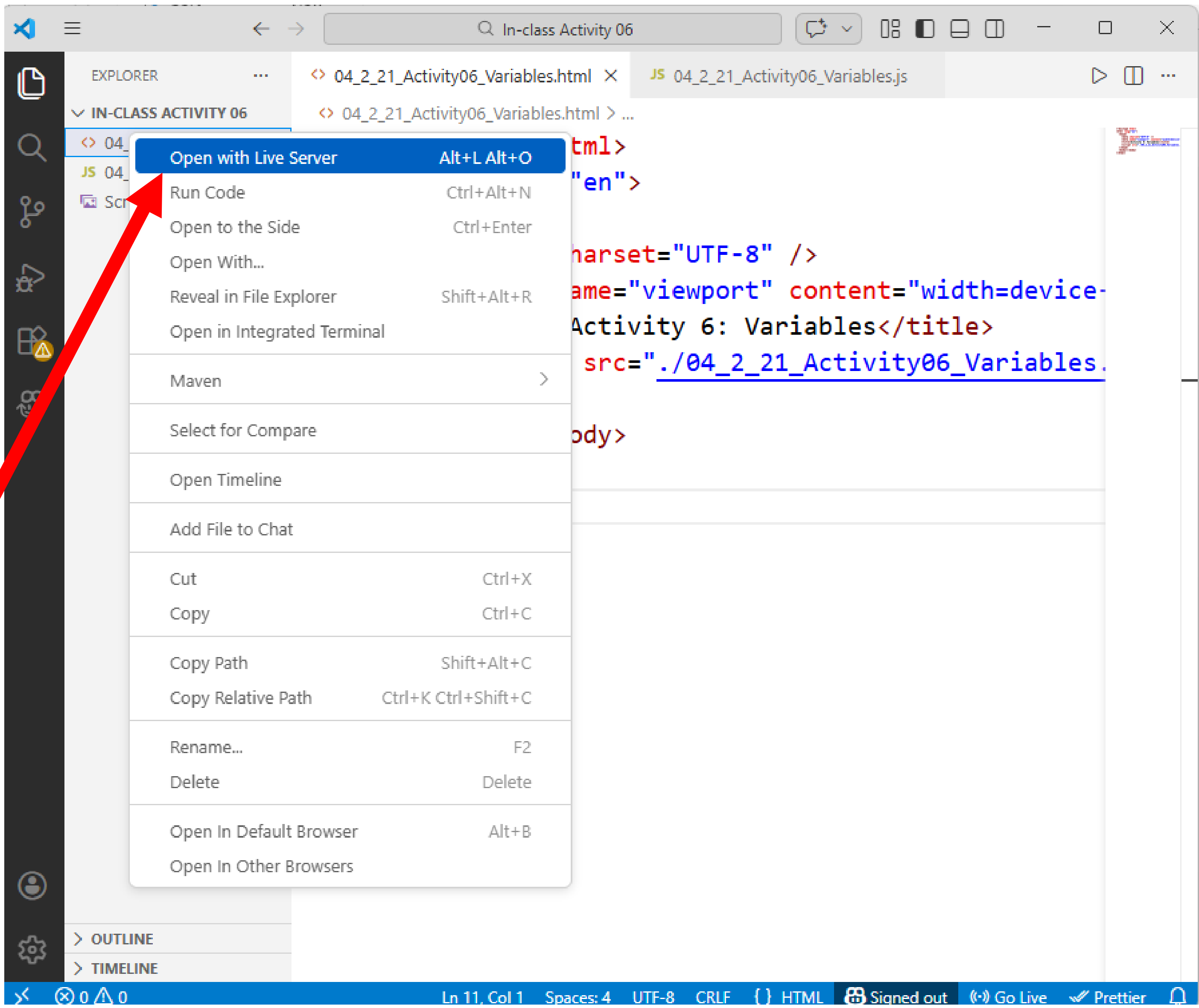
Run Live Server (Web server):





Live Server

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Variables

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In the JavaScript:

`netId_Activity06_variables.js`

Add your contact info and name of the Activity :

```
/*
Name
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Activity06 - Variables
*/
```

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```
console.log("---- I am in V A R I A B L E S ----")
```

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Data types in JavaScript

Primitive/ Value

- String
- Number
- Boolean
- Undefined   absence of a value
- null   absence of an object

Reference :

- Object
- Array
- Function

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Before 2015, there was only the keyword “`var`” in JavaScript to define variables.

2015 ES6 (ECMAScript 6) was created to standardize JavaScript.

And introduced `let` and `const`.

Let’s see the difference between `var`, `let`, and `const`.

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Write the next JavaScript :

```
// Q1 : Is it permitted the next ?

console.log("Q1 ---------------")
var var1 = "Iowa";
console.log(var1);

var var1 = 124;
console.log(var1);

// Is it permitted ?
console.log("Yes or Not");
```

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```
// Q2 : Is it valid ?
console.log("Q2 ----------------");
let var2 = "Ames";
console.log(var2);
let var2 = 124;


// Is it valid ?
console.log("Yes or Not");
```

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```
// Q3 : Is it valid ?
console.log("Q3 ----------------");
let var3 = "ISU";
console.log(var3);
var3 = 2023;
console.log(var3);


console.log("Valid ?")
```

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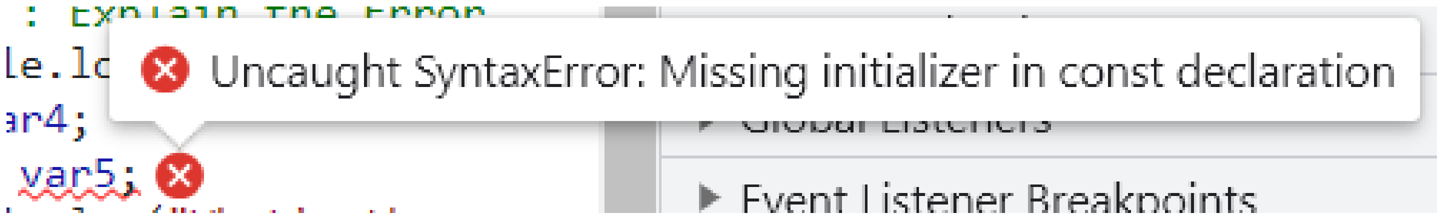
```
// Q4 : Explain the Error.
console.log("Q4 ------------------");
let var4;
const var5;


console.log("What's the error :")
```

Go to Browser>console and copy the error, then answer the question.



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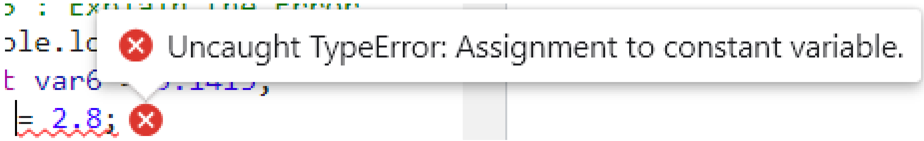
```javascript
// Q5 : Explain the Error.
console.log("Q5 -----------------");
const var6 = 3.1415;
var6 = 2.8;


console.log("What's the error :")
```

Go to Browser>console and copy the error, then answer the question.



```
// Q6 : Explain the Error.

let first name = “Sarfaraz";
console.log(" ...replace this with your response.... ");

let 2numbers = [1,2];
console.log(" ...replace this with your response.... ");

let city-state = "Ames Iowa";
console.log(" ...replace this with your response.... ");
```

```
// Q7 : What !! ??
let mainCity = "DesMoines";
console.log("This is the Capital :", MainCity)


console.log(" ....What's going on ? ....")
```

Go to Browser>console and copy the error, then answer the question.

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```javascript
// Q8 : "let" and "const" scope vs "var" scope

if (5 === 5) {
    var var7 = 100;
}
console.log(var7);

if (5 === 5) {
    let var8 = 100;
}
console.log(var8);

console.log("... explain ...")
```

Go to Browser>console and copy the error, then answer the question.

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Operators

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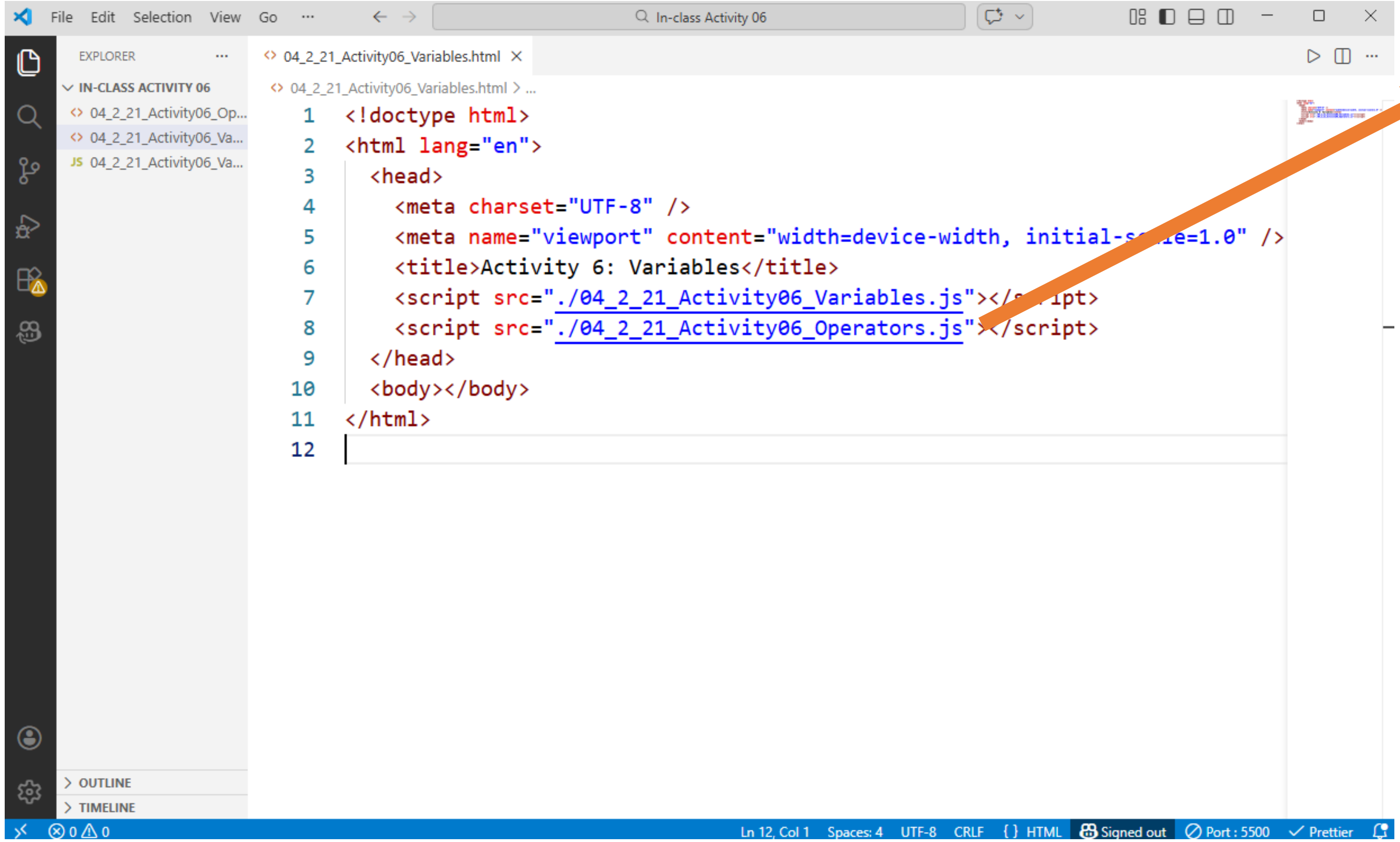
Add the JavaScript for Operators:

`netId_Activity06_index.html`

Write a JavaScript:

`netId_Activity06_operators.js`



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In the JavaScript:

`netId_Activity06_operators.js`

Add your contact info and name of the Activity :

```
/*
Name
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Activity06 - Operators
*/
```

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```
console.log("---- I am in O P E R A T O R S ----")
```

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Priority | Precedence in Arithmetic, Comparison and Boolean operators

Priority	Operator	Description
1	( )	Parenthesis
2	**	Exponentiation
3	+, -	Positive, Negative
4	*, /, %	Multiplication, Division, Modulus
5	+, -	Summation, Subtraction
6	<, <=, >, >=, !=, ==	Comparison operators
7	!	Boolean operator Not
8	&&	Boolean operator And
9	||	Boolean operator Or

Warning! These are not summation, subtraction

Higher

Precedence

Lower

Reference:
https://docs.python.org/3/reference/expressions.html#operator-precedence

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```
// Q1 : 5 + 9 / 2 + 1
```

Write a `console.log` with the response of the arithmetic expression.

Test `9/2`

```
// Q2 : 10 / 4 + 7 % 4
```

Test `10/4`

Test `7 % 4`

```
// Q3 : (5 - 1) * ((7 + 1) / (3 - 1))
```

Observe the precedence given by parenthesis ( )

```
// Q4 : 10 / 4 >= 10 / 3
```

What type of operator is >= ?

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```
// Q5 : 10 >= 10 && 10 <= 10
```

What type of operator is `&&` ?

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Equality Operators:

```
// Q6: Strict Equality
// (Type AND Value)

console.log(1 === 1)            true

console.log('1' === 1)          false
```

This is because the comparison between **`string`** and **`integer`**.

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Equality Operators:

```
// Q7 : Loose Equality
console.log('1' == 1)            true

console.log(true == 1)           true
```

This is because the comparison between **`string`** and **`integer`** It is true in lose equality.

`boolean` and **`integer`** It is true in lose equality.

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Arrays

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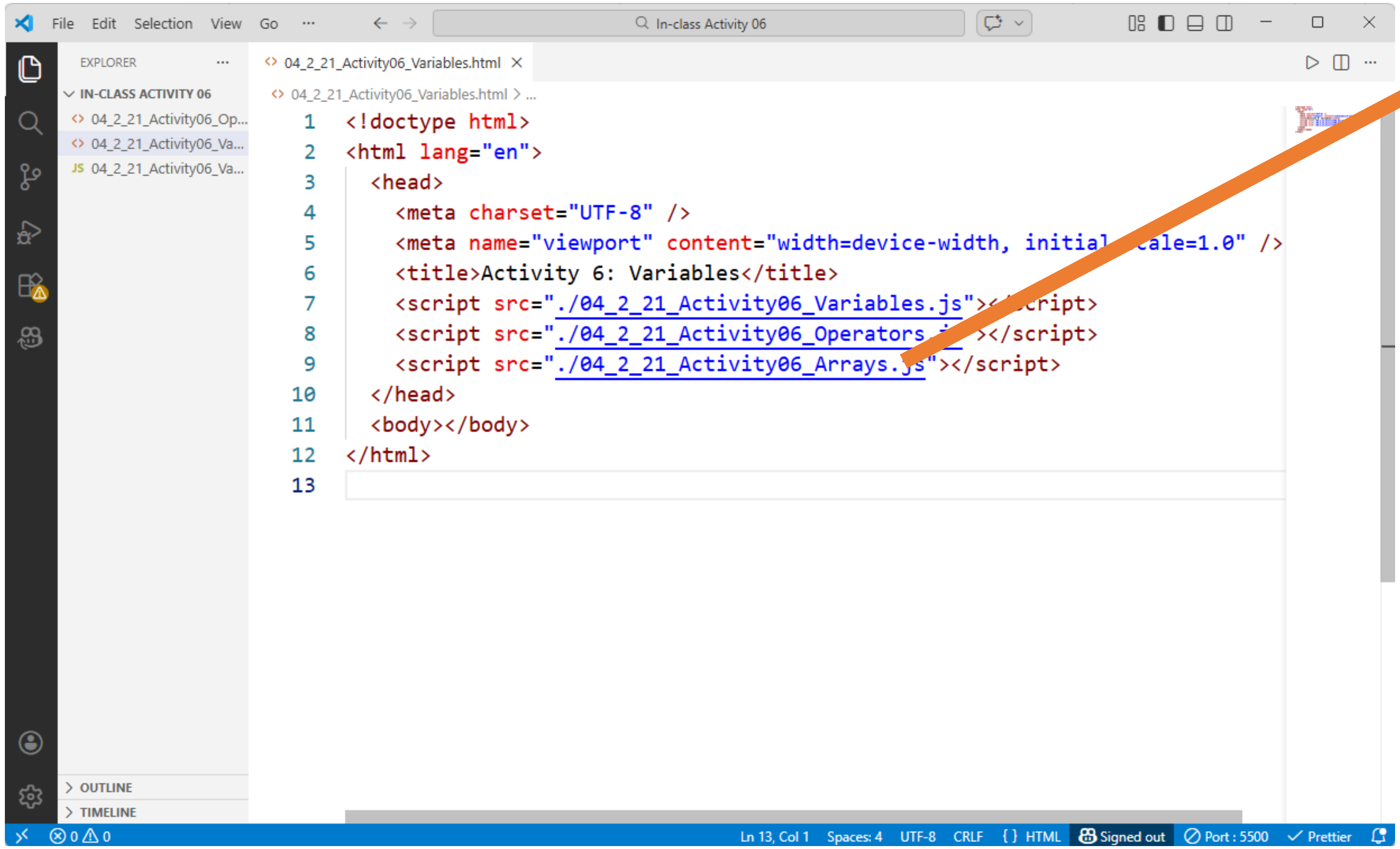
Add the JavaScript for Arrays:

`netId_Activity06_index.html`



Write a JavaScript:

`netId_Activity06_arrays.js`

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In the JavaScript:

`netId_Activity06_arrays.js`

Add your contact info and name of the Activity :

```
/*
Name
Feb 13, 2026
Activity06 - Arrays
*/
```

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```
console.log("---- I am in A R R A Y S ----")
```

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```
// Q1 : Index of each color

 let colors = ['red', 'blue', 'orange']


 e.g., console.log(colors[0]);
```

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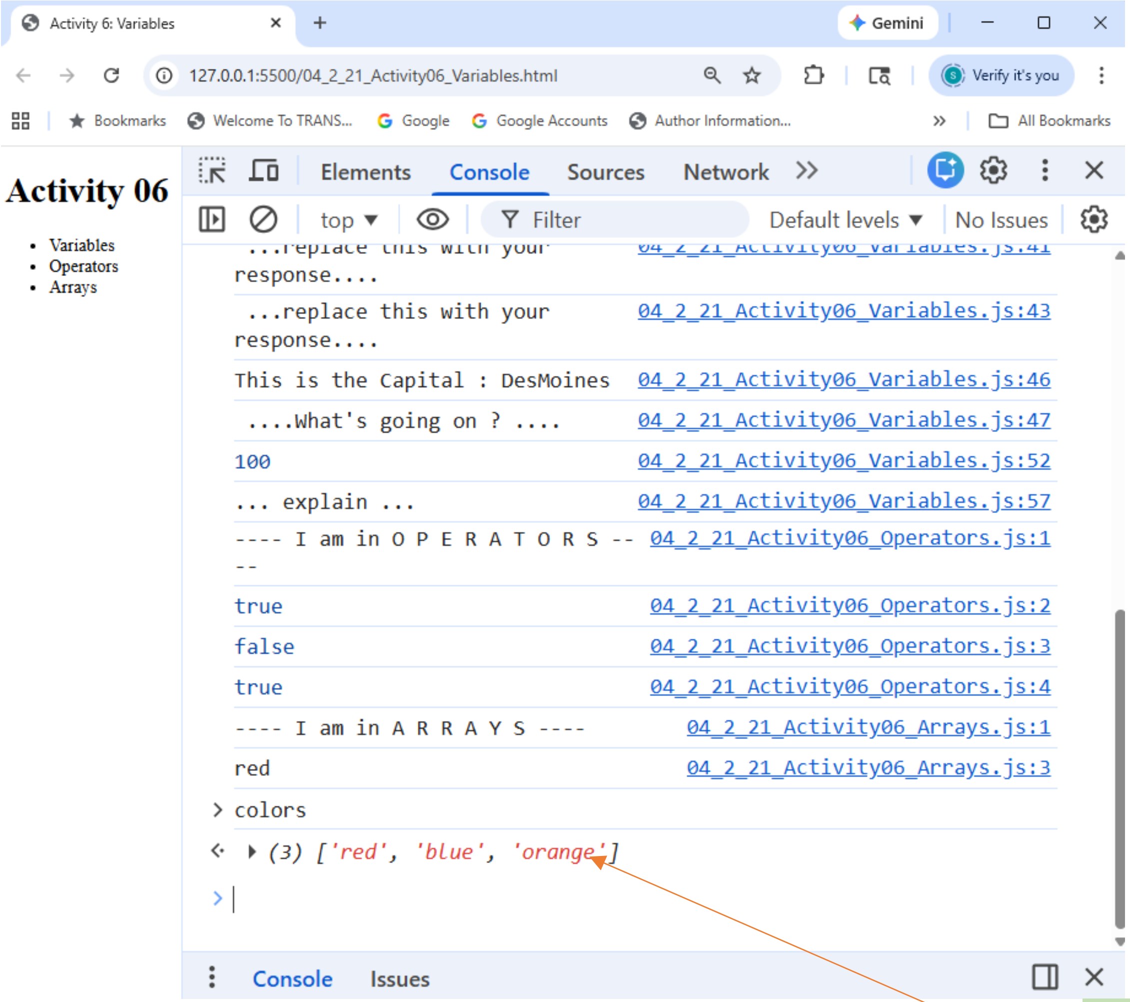

Go to the Browser > console :

Type `colors`, the variable is in memory. The Browser recognize the value `colors`.

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```
// Q2 : Add a new element 'green'

colors[3] = 'green';
console.log(colors);
```

Do you remember `colors.push()`?

What it means colors[3] ?

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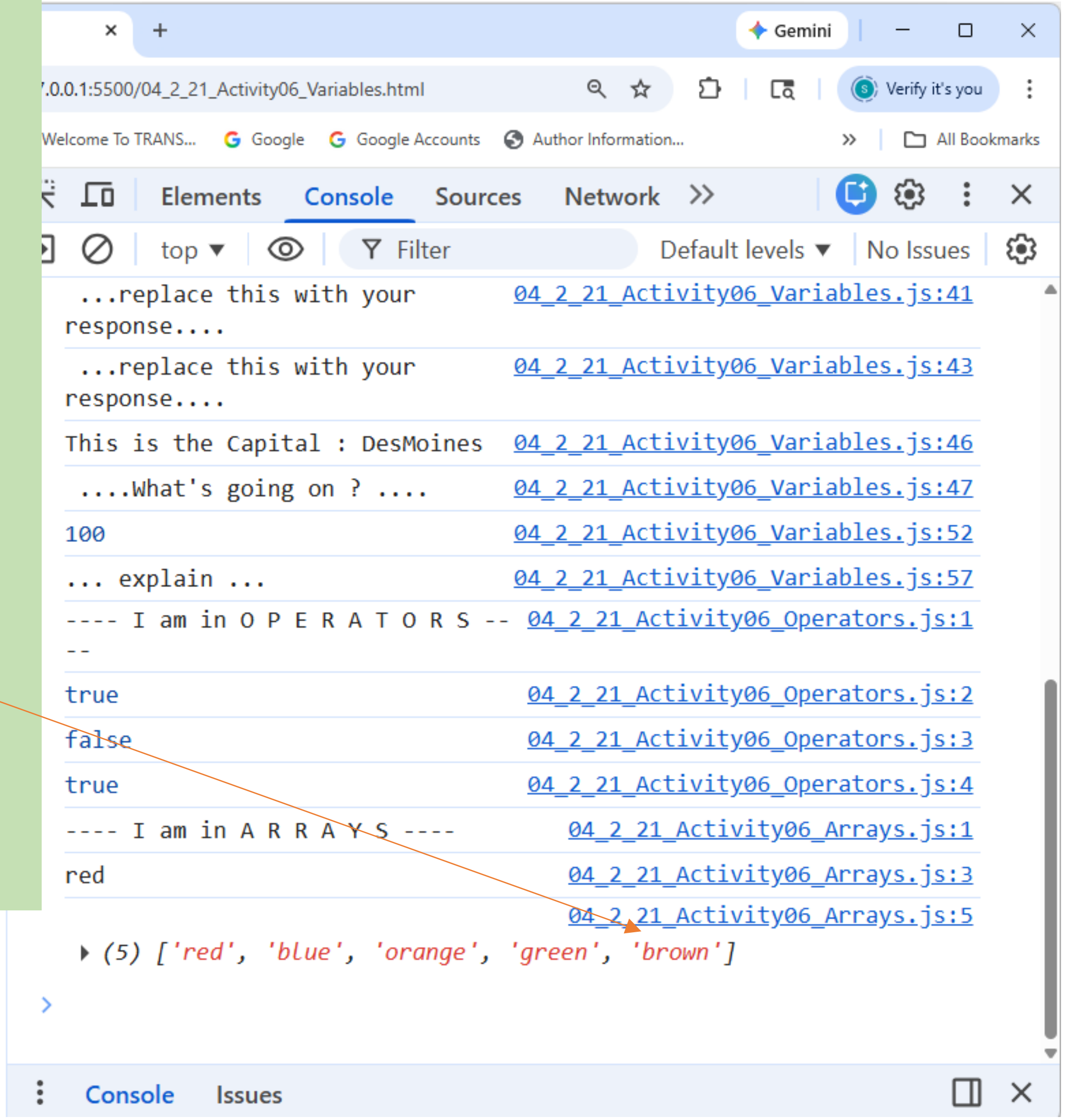
Test the next :

1. In the Activity6_arrays.js add a new color :

```
colors[4]='Brown';
```

2. Do not write `console.log(colors)` in the JavaScript file.

3. Goto to Browser, then console :
   write `colors`.



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```
// Q3 : Length of the Array

    console.log(colors.length);
```

In VSC write `console.log(colors. );`

Type `Ctrl-space`

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